

Multiple Choice

1. **Answer:** A

Options: A) 0.0125 equiv/liter; B) 0.060 equiv/liter; C) 0.050 equiv/liter; D) 0.100 equiv/liter; E) 0.125 equiv/liter

Note: The normality of the sulfuric acid solution is 0.0125 equiv/liter because sulfuric acid (H₂SO₄) is a diprotic acid that can donate two protons, meaning the equivalents of acid per mole are twice the molarity, calculated from the mass (2.45 g) divided by its molar mass (98.1 g/mol) and adjusted for the volume (2.00 L).

2. **Answer:** E

Options: A) 7.39×10^{-5} ; B) 9.12×10^{-5} ; C) 3.43×10^{-5} ; D) 6.28×10^{-5} ; E) 1.71×10^{-5}

Note: The correct answer, 1.71×10^{-5} , is derived from the Boltzmann distribution that describes the population of particles in different energy states.

2. **Answer:** E

Options: A) $8.110^6 \text{ m} \cdot \text{s}^{-1}$; B) $9.410^7 \text{ m} \cdot \text{s}^{-1}$; C) $6.210^7 \text{ m} \cdot \text{s}^{-1}$; D) $1.110^8 \text{ m} \cdot \text{s}^{-1}$; E) $3.710^7 \text{ m} \cdot \text{s}^{-1}$

Note: The correct answer is derived from the Heisenberg Uncertainty Principle, which states that the product of the uncertainties in position (x) and momentum (p) of a particle is bounded by a constant, leading to the calculation that a precise localization of an electron within 20 pm results in a corresponding uncertainty in its speed on the order of $3.7 \times 10^7 \text{ m/s}$.

3. **Answer:** A

Options: A) Bond order of He₂ is 0 and He₂⁺ is 0.5; B) Bond order of He₂ is 0.5 and He₂⁺ is 0; C) Bond order of He₂ is 0.5 and He₂⁺ is 0.5; D) Bond order of He₂ is 0 and He₂⁺ is 0.5

Note: The bond order is calculated as half the difference between the number of bonding and antibonding electrons.

3. **Answer:** A

Options: A) [H₂SO₃] > [HSO₃⁻] = [SO₃²⁻]; B) [HSO₃⁻] > [SO₃²⁻] > [H₂SO₃]; C) [SO₃²⁻] > [H₂SO₃] > [HSO₃⁻]; D) [HSO₃⁻] = [SO₃²⁻] > [H₂SO₃]; E) [SO₃²⁻] = [HSO₃⁻] > [H₂SO₃]

Note: The correct answer reflects the weak acid nature of sulfurous acid (H₂SO₃), where it only partially dissociates in solution, resulting in a higher concentration of undissociated H₂SO₃ compared to its conjugate base ions HSO₃⁻ and SO₃²⁻, which exist in equal concentrations due to the second dissociation equilibrium.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The volume of a cube is calculated by cubing the length of one edge, so for a cube with edges measuring 150.0 mm (which is 15.0 cm), the volume is $15.0\text{ cm} \times 15.0\text{ cm} \times 15.0\text{ cm} = 3375$ cubic centimeters, not 450 cubic centimeters.

7. **Answer:** True

Options: A) True; B) False

Note: The molecule NO_2^- exhibits sp^2 hybridization around its central nitrogen atom because it has one lone pair and forms two sigma bonds with oxygen atoms, resulting in a trigonal planar geometry that is characteristic of sp^2 hybridization.

8. **Answer:** False

Options: A) True; B) False

Note: The recoil energy of an atom during emission processes is typically much lower than 400 eV, as recoil energies are generally on the order of a few eV or less, depending on the mass of the emitting nucleus and the energy of the emitted particle.

9. **Answer:** False

Options: A) True; B) False

Note: The reduction of $\text{NO}(\text{g})$ typically results in the formation of nitrogen (N_2) or nitrous oxide (N_2O) rather than nitrogen dioxide (NO_2), which is produced through oxidation rather than reduction.

10. **Answer:** False

Options: A) True; B) False

Note: The statement is false because according to the Pauli Exclusion Principle, no two electrons in an atom can have the same set of quantum numbers, meaning they must differ in at least one quantum number to occupy the same orbital.

Long Form Response

11. **Answer:** The relationship between temperature and the standard molar enthalpy of sulfur dioxide gas can be understood through the concept of heat capacity and how enthalpy changes with temperature. As the temperature rises from 298.15 K to 1500 K, the standard molar enthalpy of sulfur dioxide increases due to the absorption of heat, which is quantified by the heat capacity at constant pressure. By calculating the change in enthalpy using appropriate heat capacity values, we find that the increase in enthalpy over this temperature range is approximately 62.2 kJ mol^{-1} . This value reflects the energy required for the gas to reach higher thermal states, highlighting the direct correlation between temperature and the enthalpy of a substance.
Note: correct because it accurately relates the increase in the standard molar enthalpy of sulfur dioxide gas to the principles of heat capacity, demonstrating how enthalpy changes with temperature through the absorption of heat over the specified temperature range.
12. **Answer:** When hydrogen sulfide (H_2S) is bubbled into a solution containing Zn^{2+} and Cu^{2+} ions at a concentration of 0.02 M, precipitation occurs based on the solubil-

ity products (K_{sp}) of the sulfides formed. The solubility product for CuS is significantly lower ($K_{sp} \approx 8.5 \times 10^{-45}$) compared to that of ZnS ($K_{sp} \approx 1.2 \times 10^{-23}$), indicating that CuS will precipitate first. As S^{2-} is introduced, the concentration of S^{2-} ions increases, and once it exceeds the solubility product of CuS, CuS will precipitate out of solution. In contrast, the precipitation of ZnS will not occur until the concentration of sulfide ions becomes much higher, due to its comparatively higher K_{sp} . Thus, in this scenario, CuS is expected to form first, highlighting the implications of differing solubility products on precipitation reactions. *Note:* correct because it accurately identifies that the lower solubility product of CuS compared to ZnS means that CuS will precipitate first as the concentration of sulfide ions increases in the solution.

End of Answer Key